

Enrichment: Family A (binary) vs Family B (dense)

PMBT, $p < 0.001$, max-300 descriptions

MH-Neuron / Modality Taxonomy

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Two metric families, reported side by side (not mixed). *Family A* — *binary* preserves the paper’s causal “hallucination-driving” meaning: POPE hallucination ΔH (false-yes on gt=no) and TriviaQA factual error rate $\Delta \text{Err}_{\text{TQA}}$. Single-batch ablation rarely flips a discrete answer, so these arrays are $\sim$ all-zero and the top-5% set is an ARGSORT tie-break — **flagged deg; not a valid ranking**, shown only for continuity. *Family B* — *dense* (soft $P(\text{yes})$; $1 - P_{\text{gold}}$) is continuous/tie-robust and is the *primary* result; it is being computed by a running ablation campaign (0/8 dense combos landed). OR>1 enriched $\uparrow$, <1 depleted $\downarrow$; **bold** = significant & $\geq 1.5 / \leq 0.67$.

Family A — POPE binary hallucination ΔH

Model	Combo	Visual	Text	Multimodal	Unknown	Guard
LLaVA-OneVision-7B	PMBT.gate	1.84 $\uparrow$	0.57 $\downarrow$	1.08 $\uparrow$	0.04 $\downarrow$	DEG
LLaVA-OneVision-7B	PMBT.gate_up	0.36 $\downarrow$	2.15 $\uparrow$	1.77 $\uparrow$	0.04 $\downarrow$	DEG
LLaVA-OneVision-7B	FT.gate	1.53 $\uparrow$	0.88 $\downarrow$	1.03 $\uparrow$	0.43 $\downarrow$	DEG
InternVL2.5-8B	PMBT.gate	0.48 $\downarrow$	1.81 $\uparrow$	0.98 $\downarrow$	22.82 $\uparrow$	DEG
InternVL2.5-8B	PMBT.gate_up	2.11 $\uparrow$	0.62 $\downarrow$	0.71 $\downarrow$	0.00 $\downarrow$	DEG
InternVL2.5-8B	FT.gate	0.62 $\downarrow$	0.87 $\downarrow$	1.40 $\uparrow$	2.07 $\uparrow$	DEG
Qwen2.5-VL-3B	PMBT.gate	2.02 $\uparrow$	0.81 $\downarrow$	1.04 $\uparrow$	0.03 $\downarrow$	DEG
Qwen2.5-VL-3B	PMBT.gate_up	1.38 $\uparrow$	0.91 $\downarrow$	0.93 $\downarrow$	0.02 $\downarrow$	DEG
Qwen2.5-VL-3B	FT.gate	1.18 $\uparrow$	0.78 $\downarrow$	0.93 $\downarrow$	1.06 $\uparrow$	DEG
LLaVA-1.5-7B	PMBT.gate	0.93 $\downarrow$	1.12 $\uparrow$	0.87 $\downarrow$	0.00 $\downarrow$	DEG
LLaVA-1.5-7B	PMBT.gate_up	1.04 $\uparrow$	1.05 $\uparrow$	0.80 $\downarrow$	0.25 $\downarrow$	DEG

Family A — TriviaQA binary factual error $\Delta \text{Err}_{\text{TQA}}$

Model	Combo	Visual	Text	Multimodal	Unknown	Guard
LLaVA-OneVision-7B	PMBT.gate	1.56 $\uparrow$	0.61 $\downarrow$	1.32 $\uparrow$	0.10 $\downarrow$	DEG
LLaVA-OneVision-7B	PMBT.gate_up	0.94 $\downarrow$	1.01 $\uparrow$	1.15 $\uparrow$	0.62 $\downarrow$	DEG
LLaVA-OneVision-7B	FT.gate	1.53 $\uparrow$	0.79 $\downarrow$	0.88 $\downarrow$	0.63 $\downarrow$	DEG
InternVL2.5-8B	PMBT.gate	1.52 $\uparrow$	0.74 $\downarrow$	0.83 $\downarrow$	0.00 $\downarrow$	DEG
InternVL2.5-8B	PMBT.gate_up	1.46 $\uparrow$	0.82 $\downarrow$	0.82 $\downarrow$	0.21 $\downarrow$	DEG
InternVL2.5-8B	FT.gate	1.29 $\uparrow$	0.69 $\downarrow$	1.02 $\uparrow$	0.27 $\downarrow$	DEG
Qwen2.5-VL-3B	PMBT.gate	0.88 $\downarrow$	1.21 $\uparrow$	0.85 $\downarrow$	1.03 $\uparrow$	DEG

(LLaVA-1.5 omitted from TQA — text-only TriviaQA is invalid for that backend.)

Family B — DENSE (POPE soft $P(\text{yes})$ / TQA $1 - P_{\text{gold}}$)

Pending the running dense campaign (... /dense/ per-layer ablation $\rightarrow$ 10C_*_dn enrichment). Validated live: dense ΔH is continuous and distinct per batch (40 distinct values, both signs, 0 zeros) — i.e. the guard will read *not* degenerate. Re-run this script to populate the dense tables once the results land.

TriviaQA sample-size & pool composition (spec output #4)

Effective n : POPE scores gt=no only; TQA scores the full pool. The TQA pool is $\sim$ half POPE’s and coarsely quantized (step = $1/n$), so a flagged binary TQA side is “ranking probably invalid”, not “slightly noisy”.

Model	POPE n (gt=no)	TQA n	TQA corr/incorr	TQA bin step
LLaVA-OneVision-7B	515	211	100/111	0.0047
InternVL2.5-8B	454	261	181/80	0.0038
Qwen2.5-VL-3B	497	248	82/166	0.0040
LLaVA-1.5-7B	434	274	202/72	0.0036